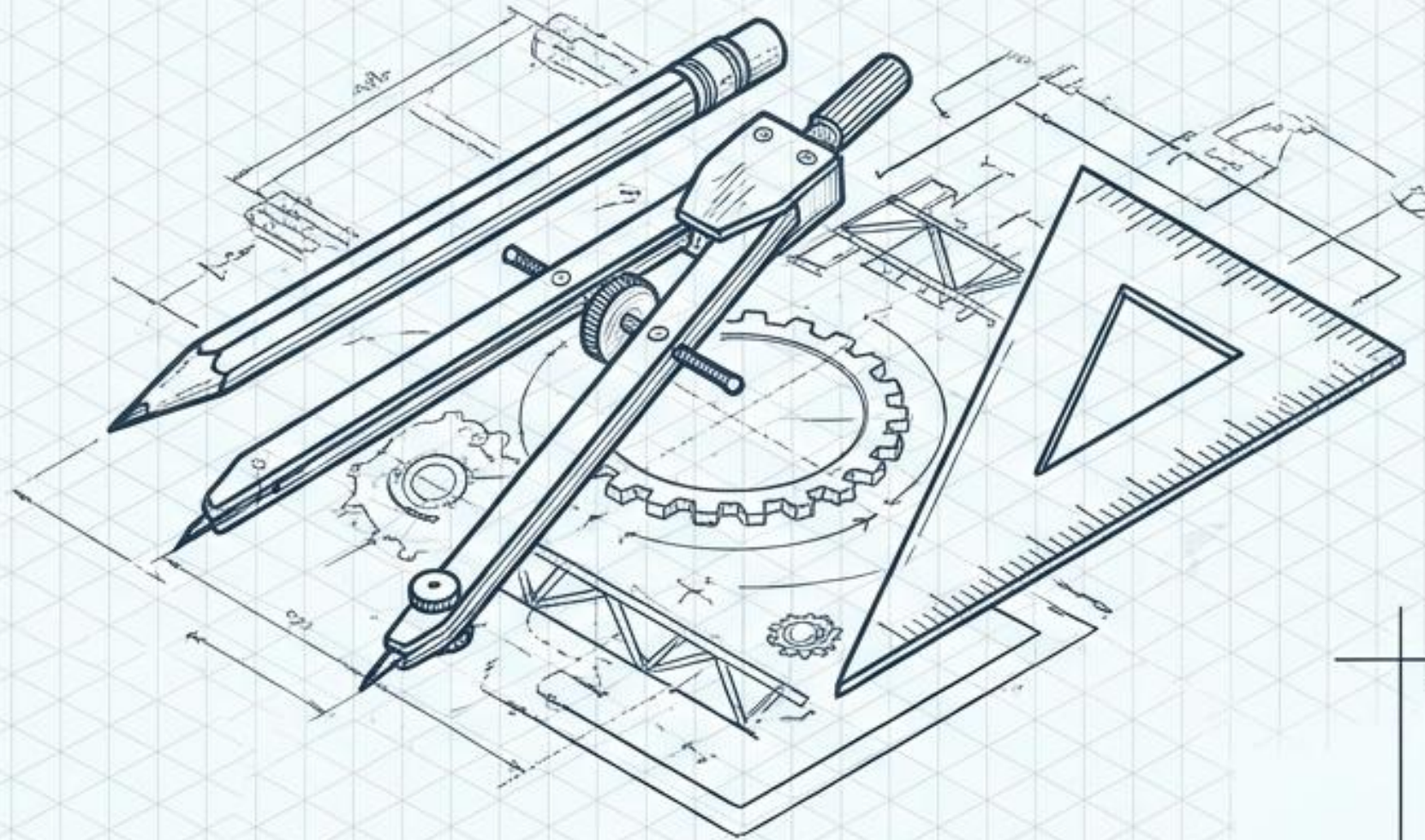


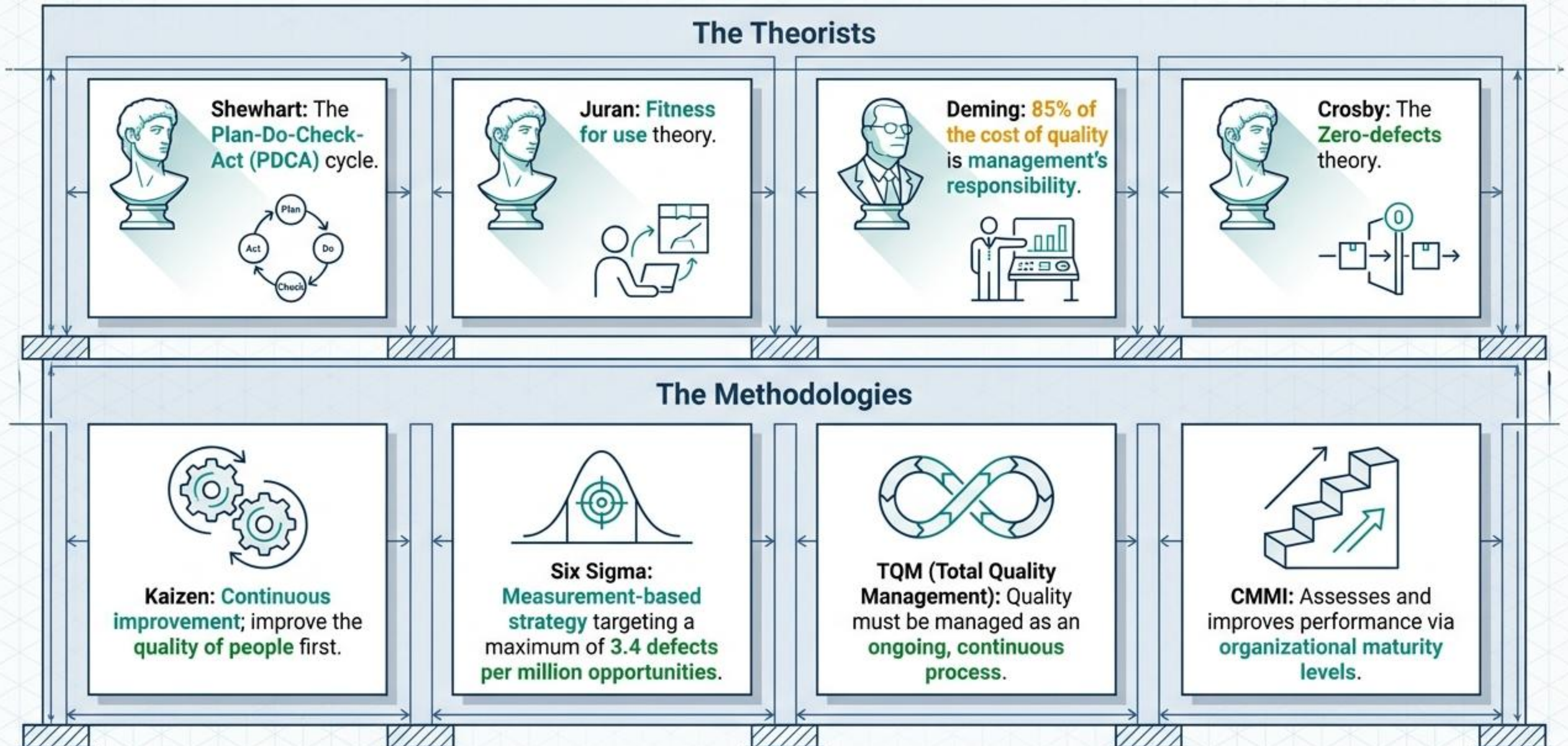
Certyfikat PMP: Project Quality Tools

The Master Craftsman's Guide to Quality Planning & Control



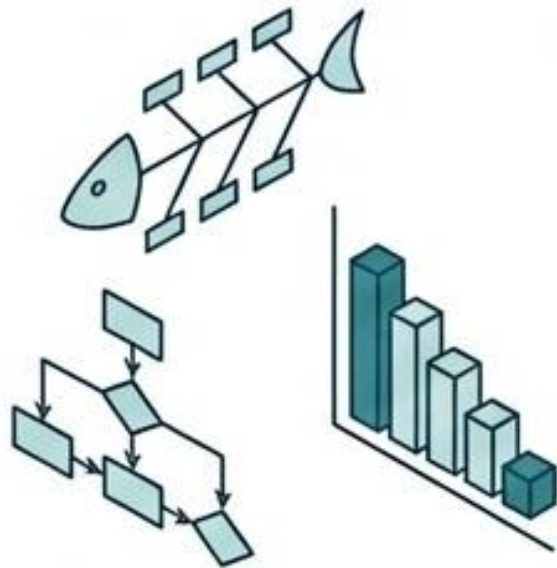
dr Zbigniew Galar

The Foundational Philosophies of Quality



The Quality Manager's Diagnostic Toolkit

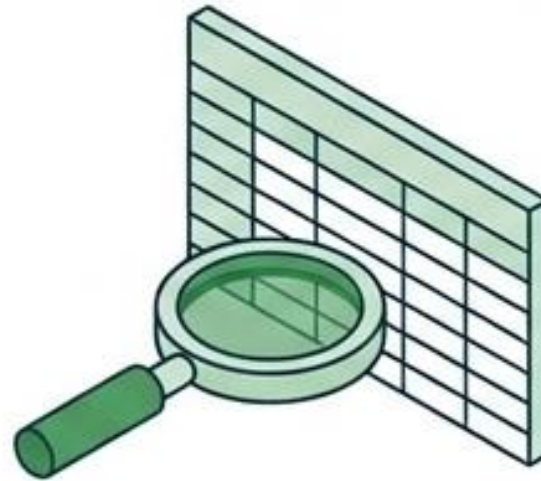
The 7 Basic Tools (Diagnostics & Graphics)



Visual mapping of processes and root cause identification.

- Ishikawa
- Flowcharts
- Checksheets
- Pareto
- Pareto
- Histograms
- Control Charts
- Scatter Diagrams

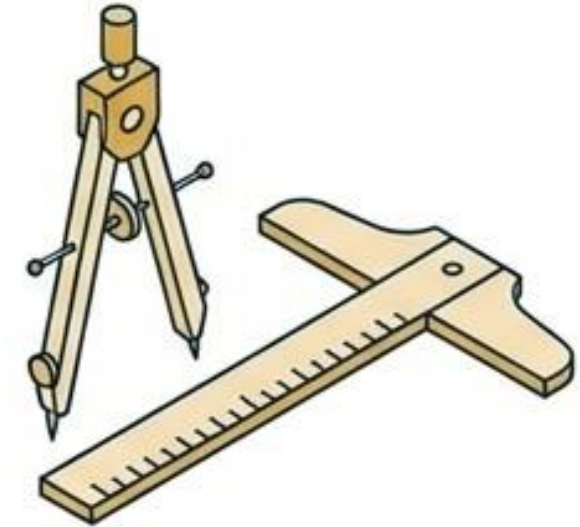
Statistical & Analytical Tools



Data-driven frameworks for variable testing and variance measurement.

- Design of Experiments (DOE)
- Benchmarking
- Statistical Sampling

Additional Planning Instruments



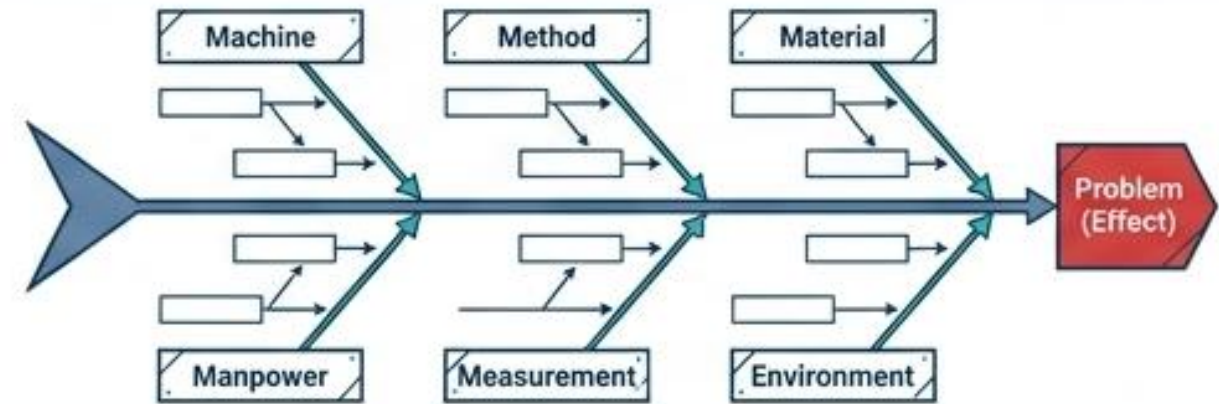
Strategic mapping for decision drivers and organizational control.

- Force Field Analysis
- Affinity Diagrams
- Matrix Diagrams

The 7 Basic Tools: Root Cause & Process Mapping

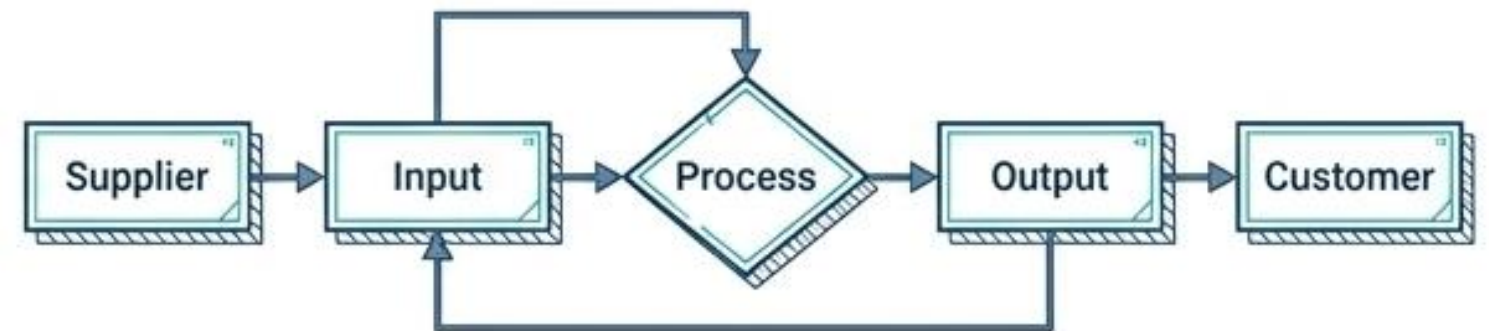
Cause-and-Effect (Ishikawa/Fishbone)

Associates the source of a problem back to a specific root cause.



Flowcharts (Stratification Charts)

Depicts relationships, decision points, and the order of steps. Helps estimate the expected costs of conformance and nonconformance.



Checksheets

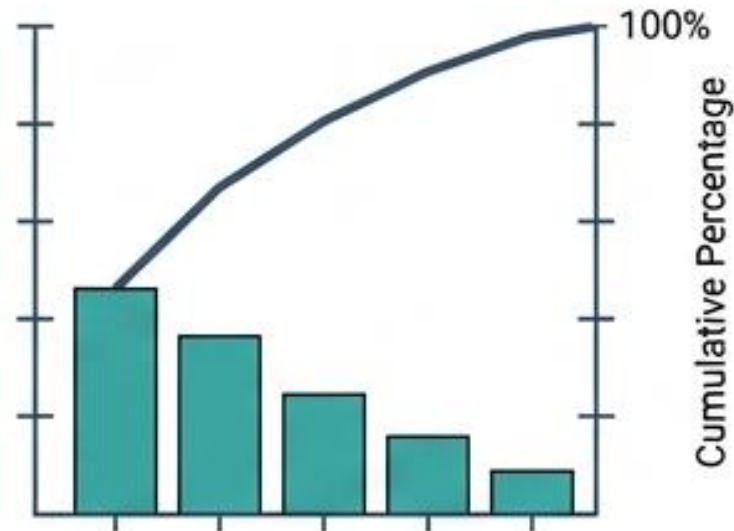
Used during quality processes to actively gather or verify data regarding quality problems during inspection.



The 7 Basic Tools: Measurement & Control Analytics

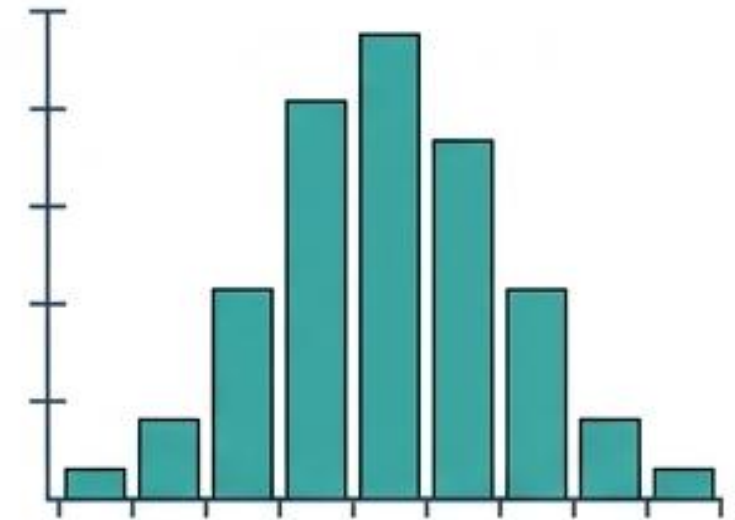
Pareto Diagrams

Identifying the vital few sources causing the majority of problems.



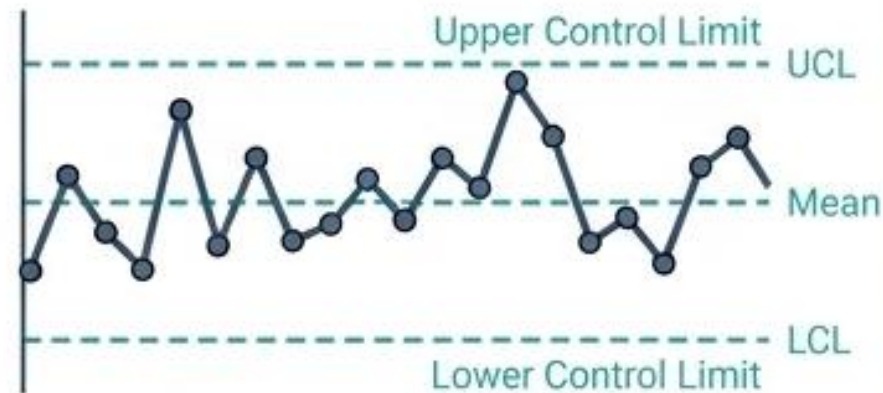
Histograms

Visualizing the shape and spread of sample data sets.



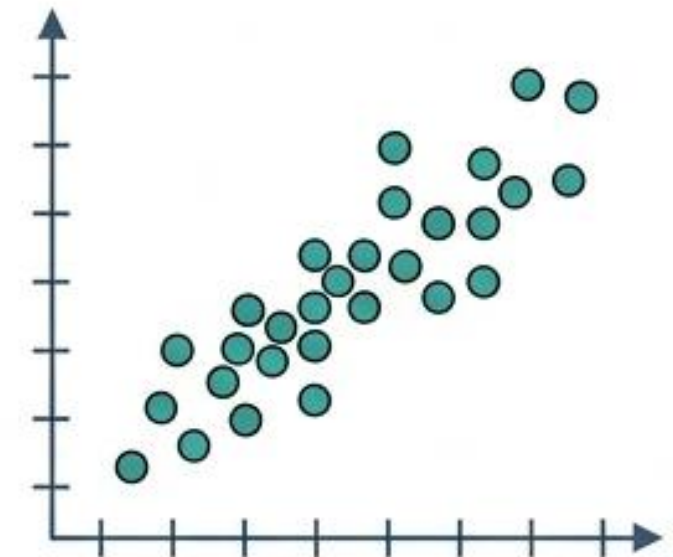
Control Charts

Monitoring process stability and detecting out-of-control variations over time.



Scatter Diagrams

Determining the relationship or correlation between two different variables.



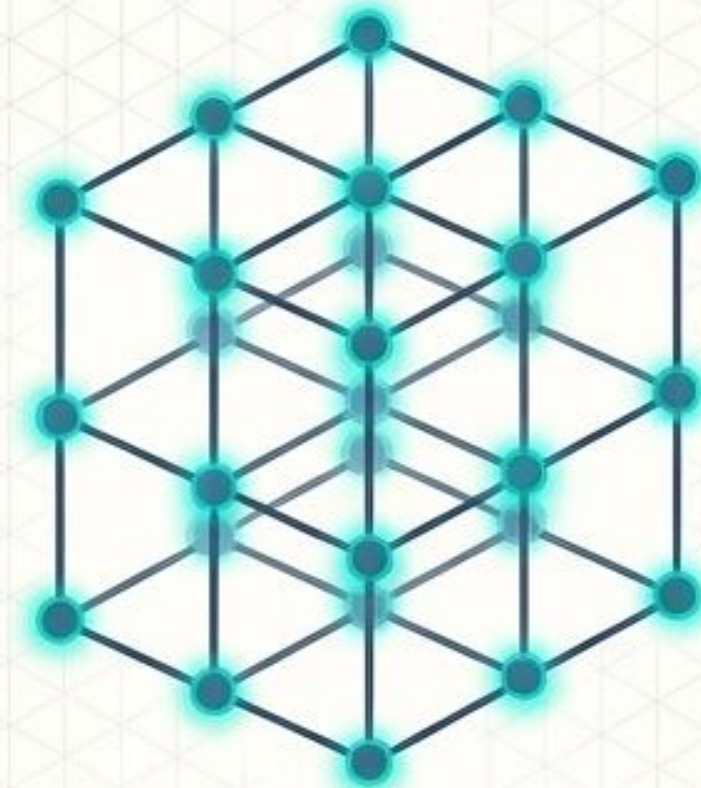
Advanced Analytics: Design of Experiments (DOE)

The Old Way (Linear)



- Testing one variable at a time.
- Slow, expensive, misses variable interactions.

The DOE Way (Statistical Matrix)



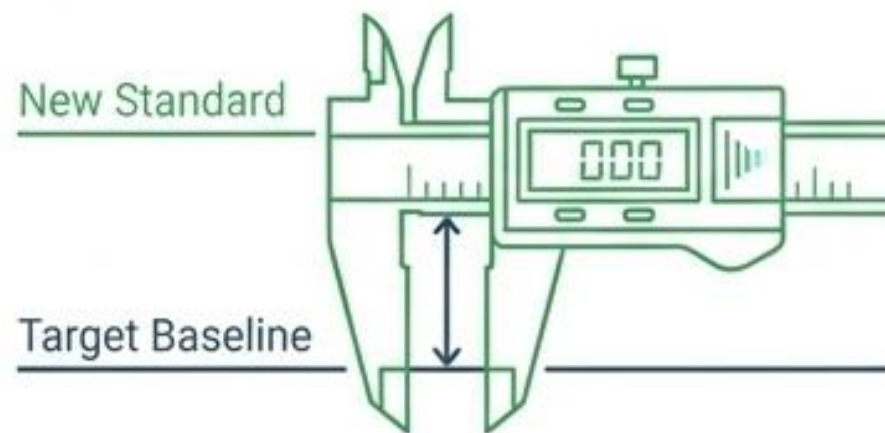
- Identifies elements/variables with the greatest effect on project outcomes.
- Determines ideal solutions using a limited number of sample cases at a reasonable cost.
- Can also be applied to project management processes to examine trade-offs.

PMP Exam Spotlight

DOE equips you with a statistical framework that allows you to change all the important variables at once, rather than one variable at a time.

Comparative Measurement Techniques

Benchmarking



Definition: Comparing previous similar activities to current project activities.

Purpose: Provides a standard to measure performance against and derives ideas for quality improvements.

Example: Upgrading a printer benchmark from 8 pages/min to a new standard of 14 pages/min.

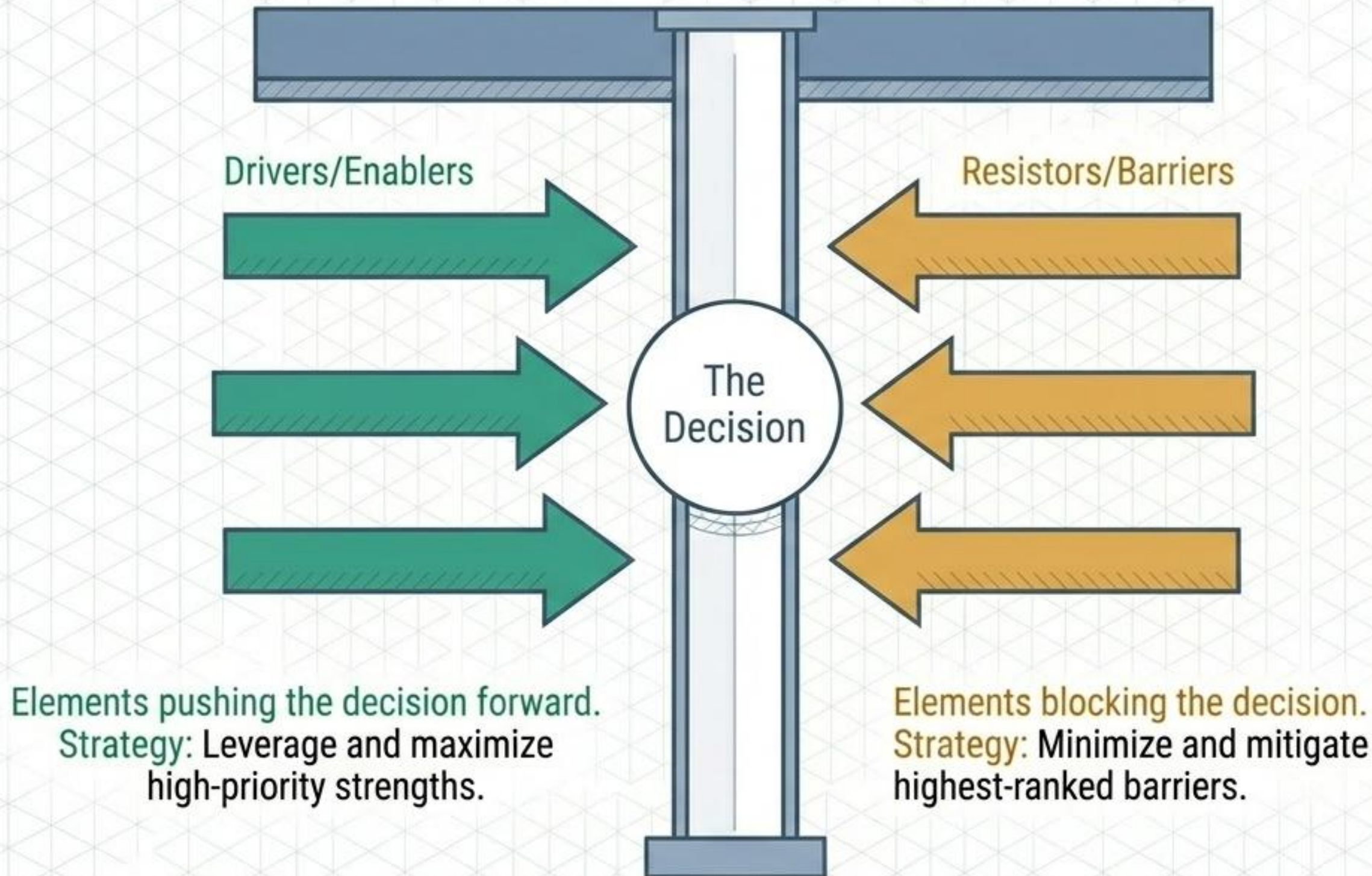
Statistical Sampling



Definition: Taking a sample number of parts from the whole population for inspection.

Purpose: Determines whether the sample falls within acceptable variances without the cost of 100% inspection.

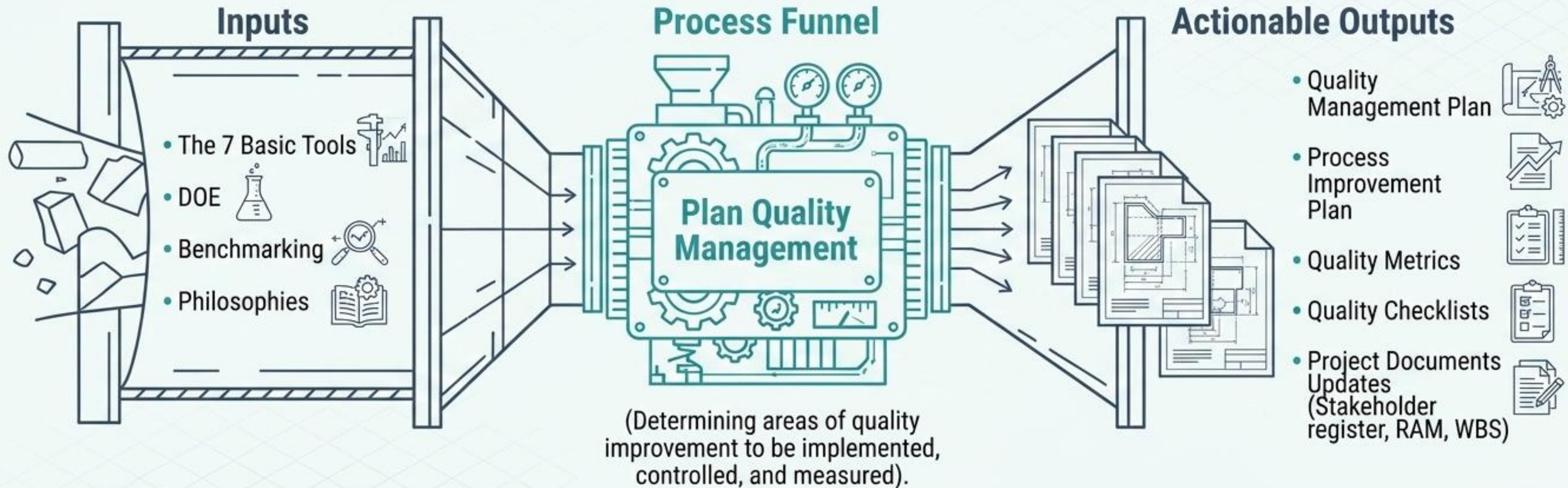
Additional Planning & Control Instruments



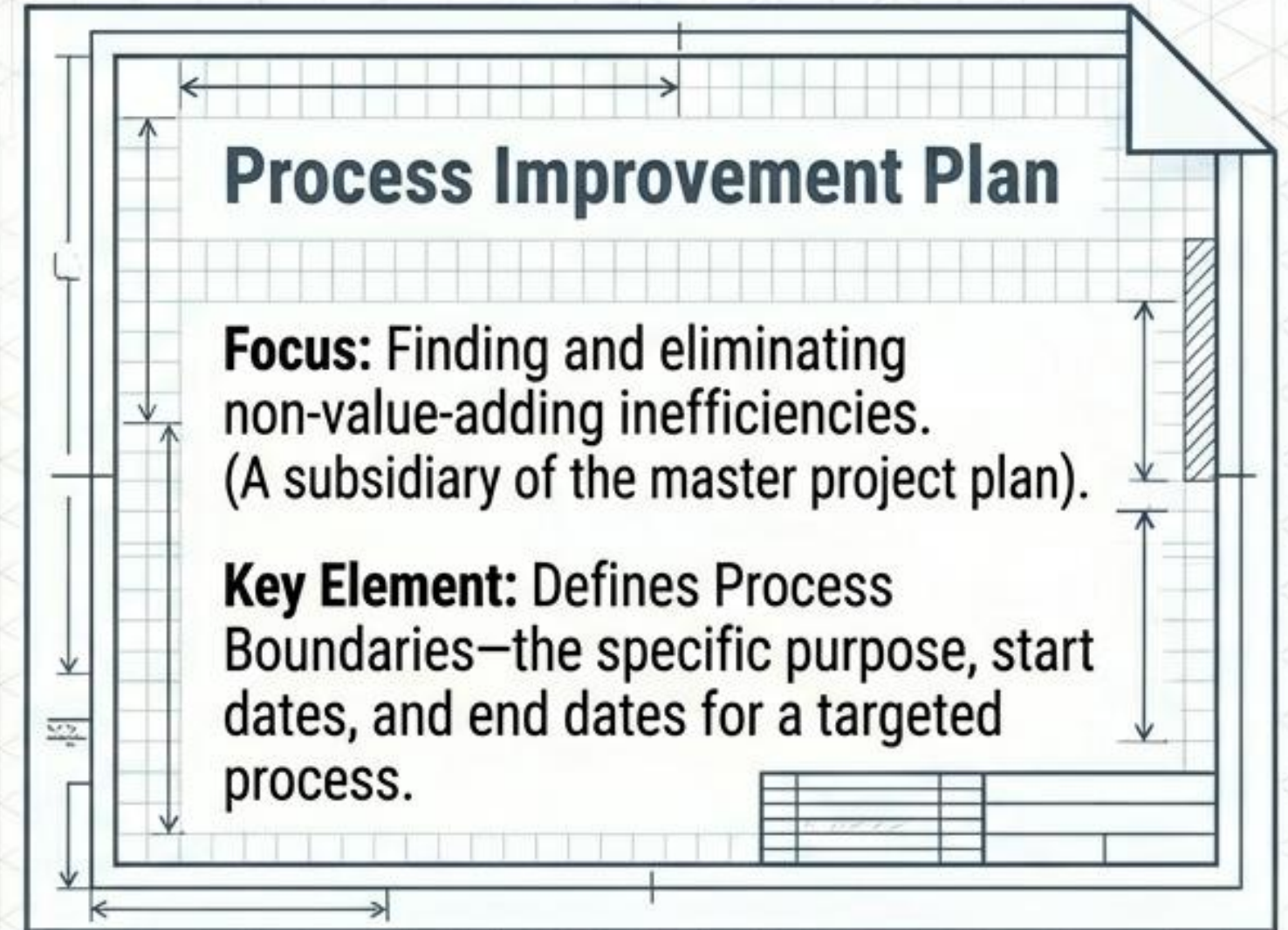
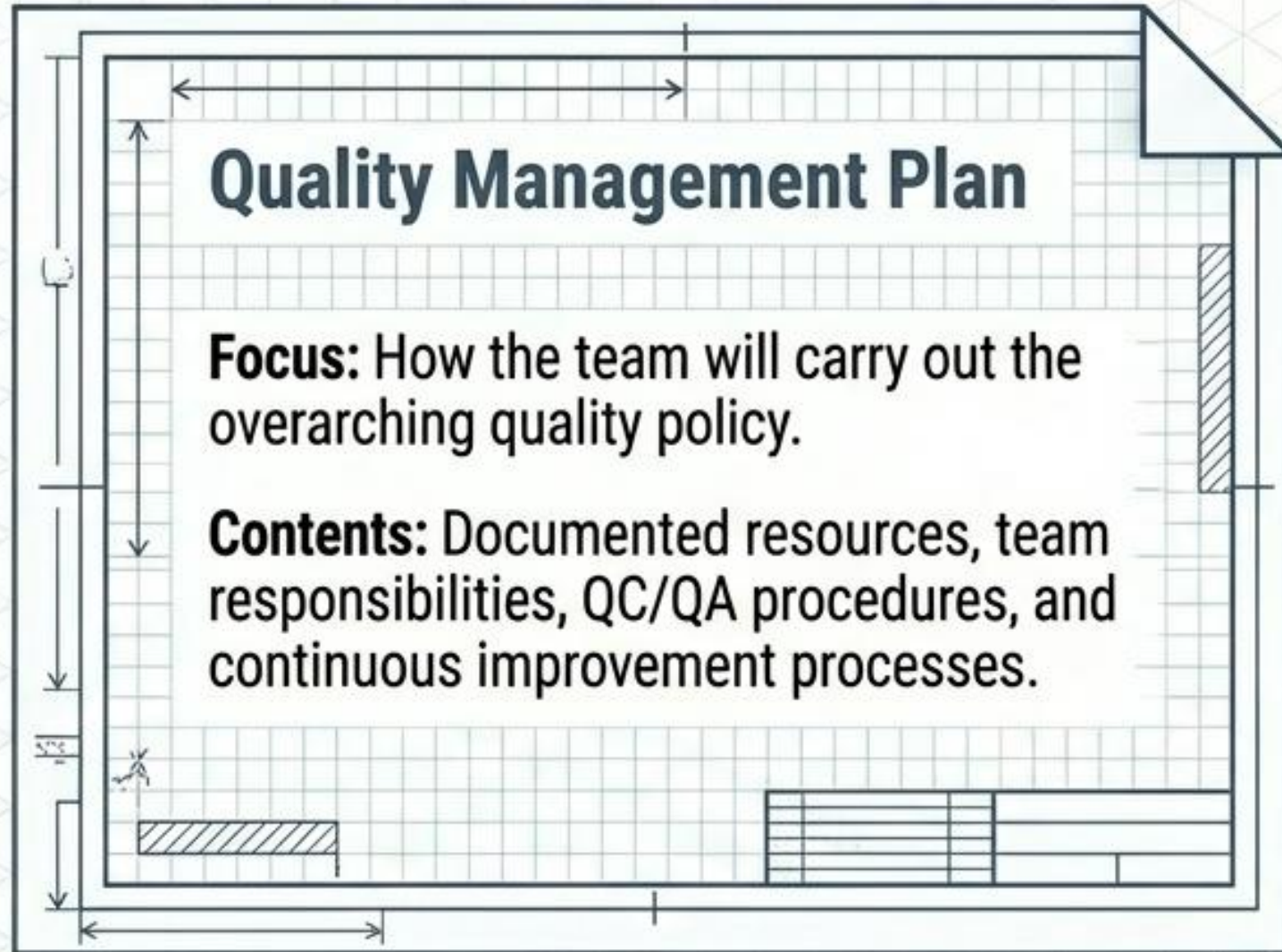
Secondary Quality Management Tools

- Brainstorming & Nominal Group Technique
- Affinity diagrams & Tree diagrams
- Process decision program charts
- Prioritization matrices & Activity network diagrams

Synthesizing the Plan: From Tools to Actionable Outputs



The Master Blueprints: Quality & Process Plans



PMP Exam Spotlight: Project Quality Management involves BOTH the management of the project process AND the quality aspects of the product/service.

Actionable Outputs: Measurement vs. Verification

Quality Metrics (Operational Definitions)



Mechanism: Describes what is measured and how during Control Quality.

Nature of Data: Consists of actual values, not just "yes/no" results (e.g., failure rates, defect density, specific delivery dates).

Quality Checklists



Mechanism: Determines whether required steps in a process have been followed.

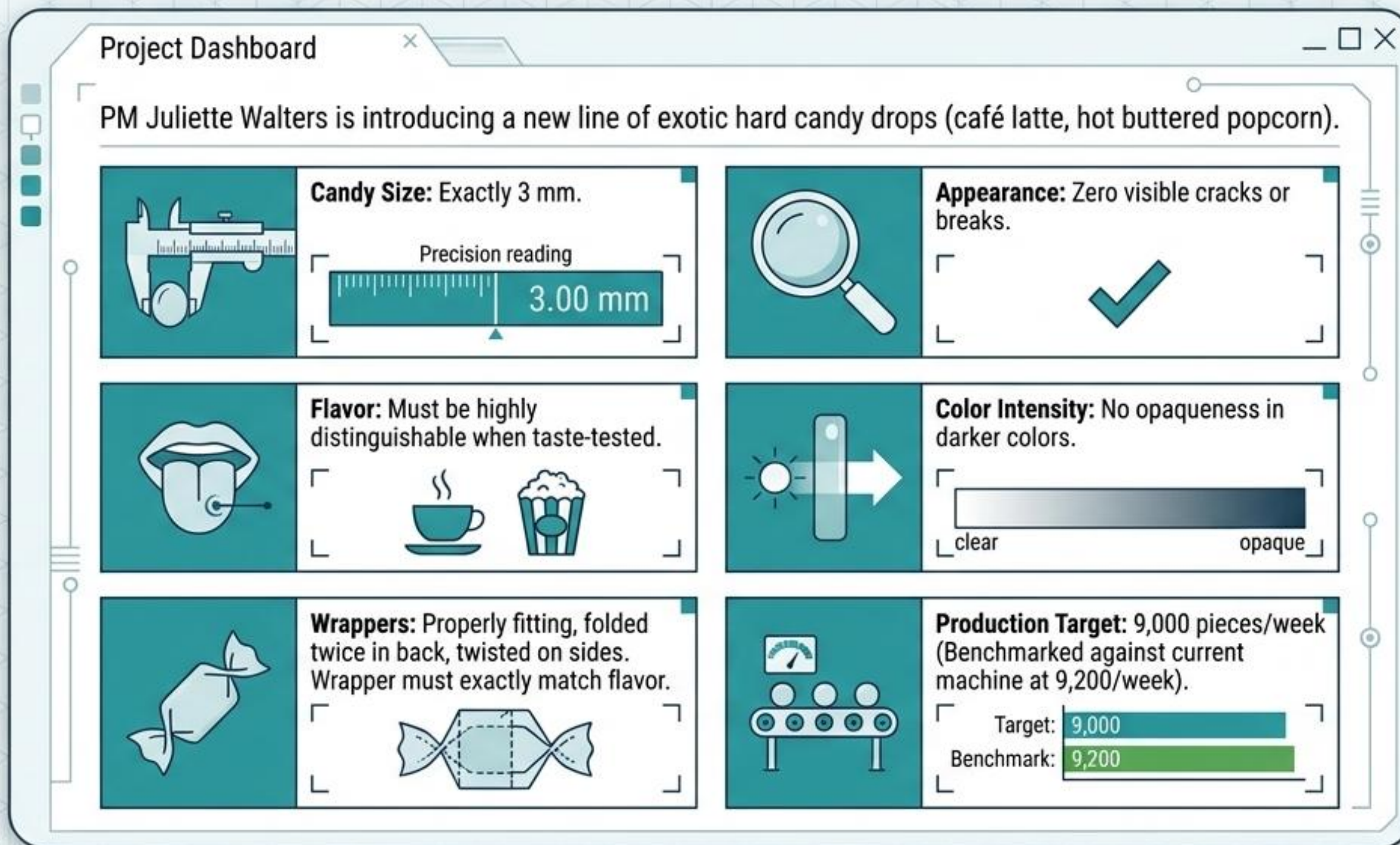
Nature of Data: Step-by-step verification; as steps complete, they are checked off.



A checklist is a multi-purpose entity: It is an output of Plan Quality Management, an input to Control Quality, and a tool (checklist analysis) in Identify Risks.

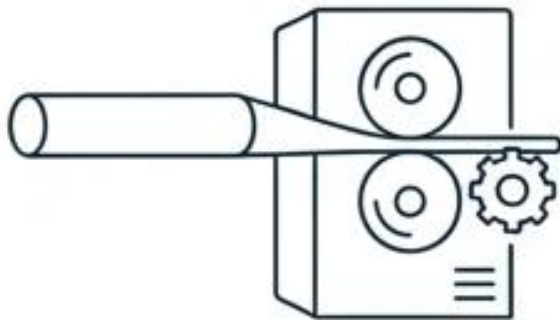


Real-World Scenario: Candy Works Factory Baseline



Solving the Quality Dilemma: The Batch Overlap

The Process



6-foot cylinders are hand-stretched and fed into a drop-maker machine.

The Point of Failure



Because cylinders vary slightly, the end of the café latte batch merges with the beginning of the hot buttered popcorn batch. Drops intermingle in the final collection bin.

The Constraint

To hit the 9,000/week metric, machines cannot be stopped between batches.

Wrapping popcorn flavor in a latte wrapper fails the quality checklist.

The Mitigation Strategy



Juliette implements a new wrapper checklist item: drops from the intermixed bins are packaged in a Mystery Flavor wrapper. Production keeps pace, and the quality problem is fully mitigated.

Integrating Quality into the Master Project Plan



The Master Baseline

The project management plan is the formal, approved map used to track and measure performance during Executing and Controlling. It contains all subsidiary plans.

Management Controls & Sign-Off



- Performance measurement baselines should change infrequently.
- Adjustments impacting cost/schedule require formalized change management and risk analysis.
- The Sign-Off Rule: Never embark on a project without formal sign-off from the project sponsor and key stakeholders on the master plan.

PMP Exam Spotlight Digest



Takeaway 1: Design of Experiments (DOE)

Remember: DOE evaluates multiple variables simultaneously using a statistical matrix, not one variable at a time.



Takeaway 2: The Scope of Quality Management

Remember: Project Quality Management governs both the management of the project process AND the quality of the final product/service.



Takeaway 3: The Dual Nature of Checklists

Remember: Quality checklists are an output of Plan Quality, an input to Control Quality, and a tool/technique in Risk Identification.



Takeaway 4: Baseline Stability

Remember: Performance measurement baselines (including the approved Project Management Plan) are management controls that should change only infrequently via formal processes.